

Task - 1: Implementation of MCP Neurons:

Design a MCP Neuron for "AND" Boolean Functions and "OR" Boolean Functions with two inputs:

```
Output of AND gate for inputs [0, 0, 1, 1] and [0, 1, 0, 1] with threshold 2: [0, 0, 0, 1]
Output of OR gate for inputs [0, 0, 1, 1] and [0, 1, 0, 1] with threshold 1: [0, 1, 1, 1]
```

Answer the Following Question:

- Question - 1: List out all the limitations of MCP - Neurons.
- Question - 2: Think if you can develop a logic to solve for XOR function using MCP Neuron. {Can you devise a if else rules.}

1. List out all the limitations of MCP - Neurons.

- MCP - Neurons cannot process non -Boolean inputs.
- All inputs are treated equally as there is no weighting mechanism.
- They fail to handle non-linearly separable functions like XOR.
- The threshold must be manually set.

2. Think if you can develop a logic to solve for XOR function using MCP Neuron. {Can you devise a if else rules.}

- Yes, an if-else rule can be designed to implement the XOR function using MCP neurons with the following logic:

XOR = False when both inputs are identical.

XOR = True when only one input is 1.

- Example: if (X1 == 1 and X2 == 0) or (X1 == 0 and X2 == 1):

result = 1

else:

result = 0

Task 2: Perceptron Algorithm for 0 vs 1 Classification.

```
Feature matrix shape: (12665, 784)
Label vector shape: (12665,)
```

Answer the Following Question:

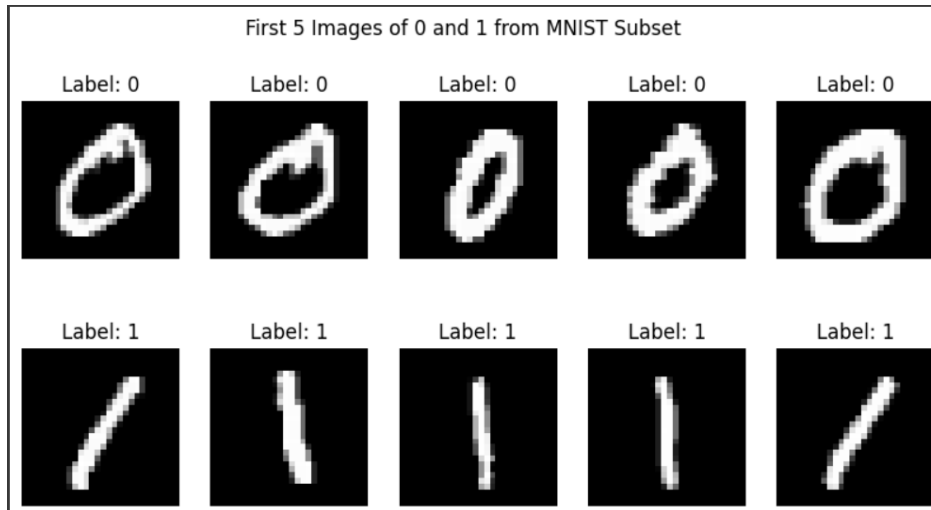
1. Question - 1: What does the shape of X represent?

- The shape of X -represents the feature matrix of given dataset, which contains the pixel values (784 pixels). The shape is (12665, 784) as each image in dataset is a 28×28 grayscale image.

2. Question - 2: What does the shape of X represent?

- The shape of Y represents the corresponding labels vector of each images in X.

Visualize the Dataset:



Question - 3: What does the weights array represent in this context?

- The weights array defines the impact of each pixel on the prediction. Each pixel's value is multiplied by its respective weight, and the sum of these products is used to generate a prediction. The array consists of 784 values, corresponding to each pixel.

Question - 4: Why are we initializing the weights to zero? What effect could this have on the training process?

- Initializing all weights to zero ensures that every feature begins with equal significance in the model. However, initializing all weights to zero can lead to identical gradient updates for each neuron in a neural network. This results in neurons learning the same features, hindering their ability to differentiate and specialize, which can negatively impact the training process.

Question - 5: What is the purpose of the output = $\text{np.dot}(X[i], \text{weights}) + \text{bias}$ line?

- The purpose of output = $\text{np.dot}(X[i], \text{weights}) + \text{bias}$ is to calculate the linear combination of the inputs and weights with bias.

Question - 6: What happens when the prediction is wrong? How are the weights and bias updated?

- When the prediction is wrong, the weights and bias are adjusted using the Perceptron Learning Rule. This update aims to reduce errors by modifying the weights and bias to improve the model's performance.

Question - 7: Why is the final accuracy important, and what do you expect it to be?

- The final accuracy is crucial as it measures how effectively the perceptron has learned from the data. For linearly separable data, 100% accuracy is expected. However, for non-linearly separable data, such as XOR, the accuracy will not reach 100% because a single-layer perceptron cannot learn such patterns.

Training the Perceptron Algorithm:

```
Epoch 90/100 - Accuracy: 100.00%
Epoch 91/100 - Accuracy: 100.00%
Epoch 92/100 - Accuracy: 100.00%
Epoch 93/100 - Accuracy: 100.00%
Epoch 94/100 - Accuracy: 100.00%
Epoch 95/100 - Accuracy: 100.00%
Epoch 96/100 - Accuracy: 100.00%
Epoch 97/100 - Accuracy: 100.00%
Epoch 98/100 - Accuracy: 100.00%
Epoch 99/100 - Accuracy: 100.00%
Epoch 100/100 - Accuracy: 100.00%
The Final Accuracy is: 100.0
```

Visualizing Misclassified Image.

Final Accuracy: 1.0000
All images were correctly classified!

Question - 8: What does misclassified_idx store, and how is it used in this code?

- The misclassified_idx stores the indices of the misclassified images, where the predicted labels do not match the actual labels. It is used to identify and track which images were incorrectly predicted during the training process.

Question - 9: How do you interpret the result if the output is "All images were correctly classified!"?

- If the output is "All images were correctly classified!", it means the perceptron has achieved 100% accuracy, indicating that all predictions matched the actual labels and no errors were made.

Task 3: Perceptron Algorithm for 3 vs 5 Classification.

Epoch 98/100 - Accuracy: 98.94%
Epoch 99/100 - Accuracy: 98.47%
Epoch 100/100 - Accuracy: 98.58%
Final Accuracy: 98.58%
Final Accuracy: 98.69%

Misclassified Images

Pred: 1, True: 0



Pred: 0, True: 1



Pred: 0, True: 1



Pred: 0, True: 1



Pred: 1, True: 0



Pred: 0, True: 1



Pred: 0, True: 1



Pred: 0, True: 1



Pred: 0, True: 1



Pred: 1, True: 0



Conclusion

The final perceptron accuracy is 98.69%, indicating that it has successfully distinguished between the digits 5 and 3 in most cases. However, some misclassified images were observed due to overlapping handwriting styles, where the digit 3 resembled 5 and vice versa. This overlap in writing patterns is the reason the accuracy did not reach 100% in this scenario.